Changshen (Tony) Chen

Los Angeles, CA(open to relocate)• +1 (626) 557-2927 • chenchangshen899@gmail.com • linkedin.com/in/changshen-chen

EDUCATION

Columbia University, Mailman School of Public Health — New York, NY
M.S. in Biostatistics — *Expected May 2028*

- Relevant coursework: Statistical Inference, Regression Analysis, Survival Analysis, Data Science I–II

University of Southern California — Los Angeles, CA
B.S. in Quantitative Biology | GPA: 3.69 / 4.00 — *May 2026*

TECHNICAL SKILLS

Programming: R (tidyverse, ggplot2, limma, DESeq2, survival), Python (pandas, NumPy, scikit-learn), Bash, C++

Biostatistics: Linear & Logistic Regression, Generalized Linear Models, Survival Analysis, Kaplan-Meier Estimation, Cox Proportional Hazards Models, Hypothesis Testing, ANOVA, Multiple Testing Adjustment, Statistical Inference

Tools & Platforms: Git/GitHub, High-Performance Computing, R Markdown, Microsoft Excel

Languages: English (fluent), Mandarin (native), French (intermediate)

PROFESSIONAL EXPERIENCE

Bristol Myers Squibb (BMS) — Shanghai, CN
Data & Quality Operations Intern, Manufacturing Division — *Jun 2024 – Jul 2024*

- Cleaned, reconciled, and analyzed **supplier, purchase-order, and quality datasets**, performing data validation, descriptive statistics, and variance analysis to identify missing, inconsistent, and anomalous records.
- Conducted microbiological and chemical assays under **GMP-controlled procedures**, validating experimental results against predefined specifications and maintaining accurate quality documentation.
- Developed **standardized data summaries and visualizations** to track quality and operational metrics, supporting cross-functional review and data-driven decision-making.

Becton, Dickinson and Company (BD) — Shanghai, CN
Supply Chain Analyst Intern — *Jul 2023 – Aug 2023*

- Cleaned and integrated **2,000+ operational records** across cargo movement, import/export, customs clearance, and warehouse inventory, identifying missing values, duplicates, and inconsistent entries.
- Performed **descriptive statistical and trend analyses on 1,000+ transactions**, developing analytical reports and visualizations to identify logistics, clearance-time, and inventory-movement patterns.
- Built **reproducible data-processing workflows in Python and Excel** to automate data transformation, aggregation, and quality checks, improving the efficiency and consistency of recurring analyses.

RESEARCH EXPERIENCE

Prediction of antibody-mediated rejection in kidney transplantation — Los Angeles, CA
Supervised by Dr. Chongming Jiang at Terasaki Institute — *Mar 2025 – Sep 2025*

- Integrated and processed **3 GEO transcriptomic datasets**, performing data cleaning, normalization, feature harmonization, and quality control in R to create analysis-ready datasets.
- Performed **differential expression, GSEA, and supervised machine-learning analyses** to identify candidate biomarkers and develop a cross-validated classifier distinguishing rejection from stable-graft patients.

Multi-Omic Profiling of Lung Adenocarcinoma — Los Angeles, CA
Undergraduate Researcher — *Jan 2024 – Dec 2024*

- Extracted, merged, and cleaned patient-level genomic and clinical data from the GDC database, addressing **missing values, duplicate records, and inconsistent clinical annotations** to support reliable downstream analyses.
- Applied **Kaplan-Meier estimation, log-rank tests, and Cox proportional hazards regression** to evaluate associations between gene expression and patient survival and identify potential prognostic biomarkers.

PUBLICATION

Chen, C. (2021). *Investigation of Potential Inhibitors of Brown Adipose Tissue Induced Thermogenesis.* In **Proceedings of the 11th International Conference on Bioscience, Biochemistry and Bioinformatics (ICBBB '21)**, Tokyo, Japan.